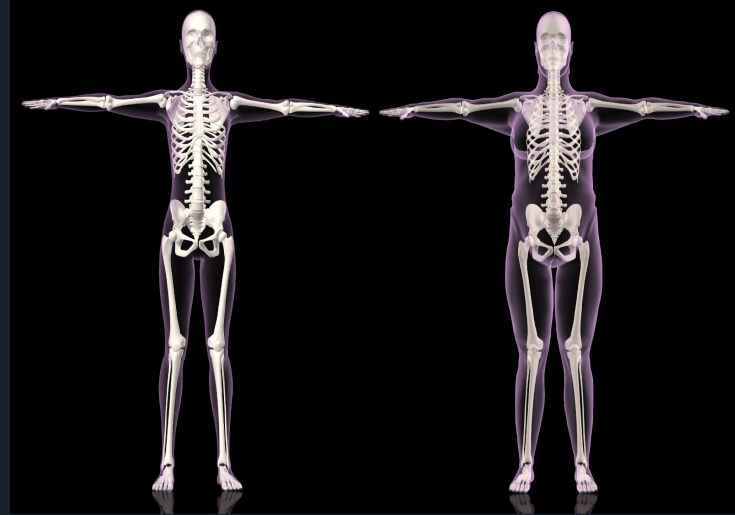


A decorative graphic on the left side of the slide consisting of two overlapping parallelograms. The front one is blue and the back one is a light green color. They are positioned diagonally, with the blue one in front of the green one.

Estimates of Obesity Levels Based on Eating Habits and Physical Conditions

Introduction

- Obesity is widely recognized as a multifactorial condition driven by the interaction of diet, physical activity, environment, and behavior.
- Numerous studies associate frequent consumption of energy-dense, high-calorie foods (such as fast food and sugary drinks) with increased BMI and higher chances of being overweight.
- The goal of this project is to investigate which lifestyle and demographic factors are most strongly associated with obesity levels.





Data Cleaning and Preparation

No missing values:

We checked every column and found zero missing entries, so the full dataset could be used without dropping or filling any cases.

For hypothesis 1 the multi-category obesity label (NObeyesdad) was recoded into a binary variable (OverObese) to make it compatible with logistic regression. High-calorie food consumption (FAVC) was converted into a simple yes/no indicator (FAVC_bin) to improve clarity and interpretability. Meal frequency (NCP) was transformed from a string-based field into a numeric variable (NCP_num) so it could be used as a continuous predictor. Together, these preprocessing steps produced a clean and consistent dataset suitable for reliable statistical analysis.

Example Table

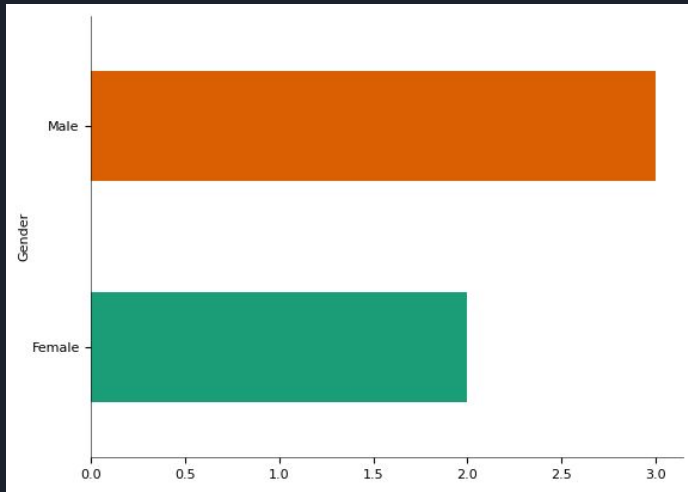
Raw Data Preview (from ucimlrepo):

	Gender	Age	Height	Weight	family_history_with_overweight	FAVC	FCVC	NCP	CAEC	SMOKE	CH2O	SCC	FAF	TUE	CALC	MTRANS	NObeyesdad
0	Female	21.0	1.62	64.0	yes	no	2.0	3.0	Sometimes	no	2.0	no	0.0	1.0	no	Public_Transportation	Normal_Weight
1	Female	21.0	1.52	56.0	yes	no	3.0	3.0	Sometimes	yes	3.0	yes	3.0	0.0	Sometimes	Public_Transportation	Normal_Weight
2	Male	23.0	1.80	77.0	yes	no	2.0	3.0	Sometimes	no	2.0	no	2.0	1.0	Frequently	Public_Transportation	Normal_Weight
3	Male	27.0	1.80	87.0	no	no	3.0	3.0	Sometimes	no	2.0	no	2.0	0.0	Frequently	Walking	Overweight_Level_I
4	Male	22.0	1.78	89.8	no	no	2.0	1.0	Sometimes	no	2.0	no	0.0	0.0	Sometimes	Public_Transportation	Overweight_Level_II

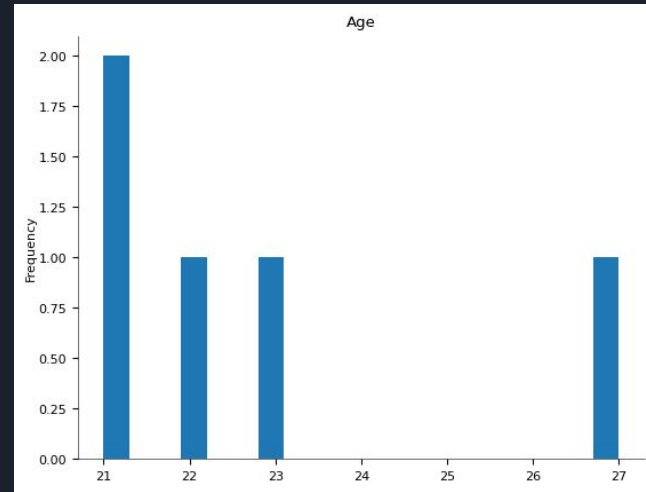
Shape: (2111, 17)

This is the first 5 values of the data set

Demographics Of Dataset



60% Males 40% Females



40% - Age 21
20% - Age 22
20% - Age 23
20% - Age 27



Hypothesis 1: Lifestyle Factors - Diet and Eating Frequency

Hypothesis 1: Individuals who frequently consume high-calorie foods and eat more meals per day are more likely to be overweight or obese.

Dataset: UCI Obesity Levels (n=2,111)

Variables:

- **OverObese** (1 = overweight/obese, 0 = normal/underweight)
- **FAVC_bin** (high-calorie consumption)
- **NCP_num** (meal frequency)
- **Interaction:** $\text{FAVC} \times \text{NCP}$



Binary Logistic Regression Model:

$$OverObese = \beta_0 + \beta_1(FAVC_{bin}) + \beta_2(NCP_{num}) + \beta_3(FAVC_{bin} \times NCP_{num}).$$

Test whether:

1. High-calorie food consumption predicts obesity
2. Meal frequency predicts obesity
3. The effect of meal frequency differs by high-calorie consumption



Binary Logistic Regression Model:

To test Hypothesis 1, we fit a logistic regression model using high-calorie food consumption (FAVC) and meal frequency (NCP) as predictors of obesity status. The model included both main effects and their interaction to assess whether the influence of meal frequency differs by high-calorie eating behavior. We evaluated predictor significance using odds ratios and p-values and assessed overall model performance using ROC/AUC metrics and predicted probability plots.



Logistic Regression Assumptions:

Logistic Regression Assumptions:

- No multicollinearity ($VIF < 5$ for all predictors)
- Appropriate functional form (S-curve check)
- Overall model fit evaluated (Pseudo R^2 , AUC)

Results: No major violations of logistic regression assumptions were detected, indicating the model is statistically sound and appropriate for inference.



Distribution of Weight Categories (N0beyesdad):

count

N0beyesdad

Obesity_Type_I	351
Obesity_Type_III	324
Obesity_Type_II	297
Overweight_Level_I	290
Overweight_Level_II	290
Normal_Weight	287
Insufficient_Weight	272

dtype: int64

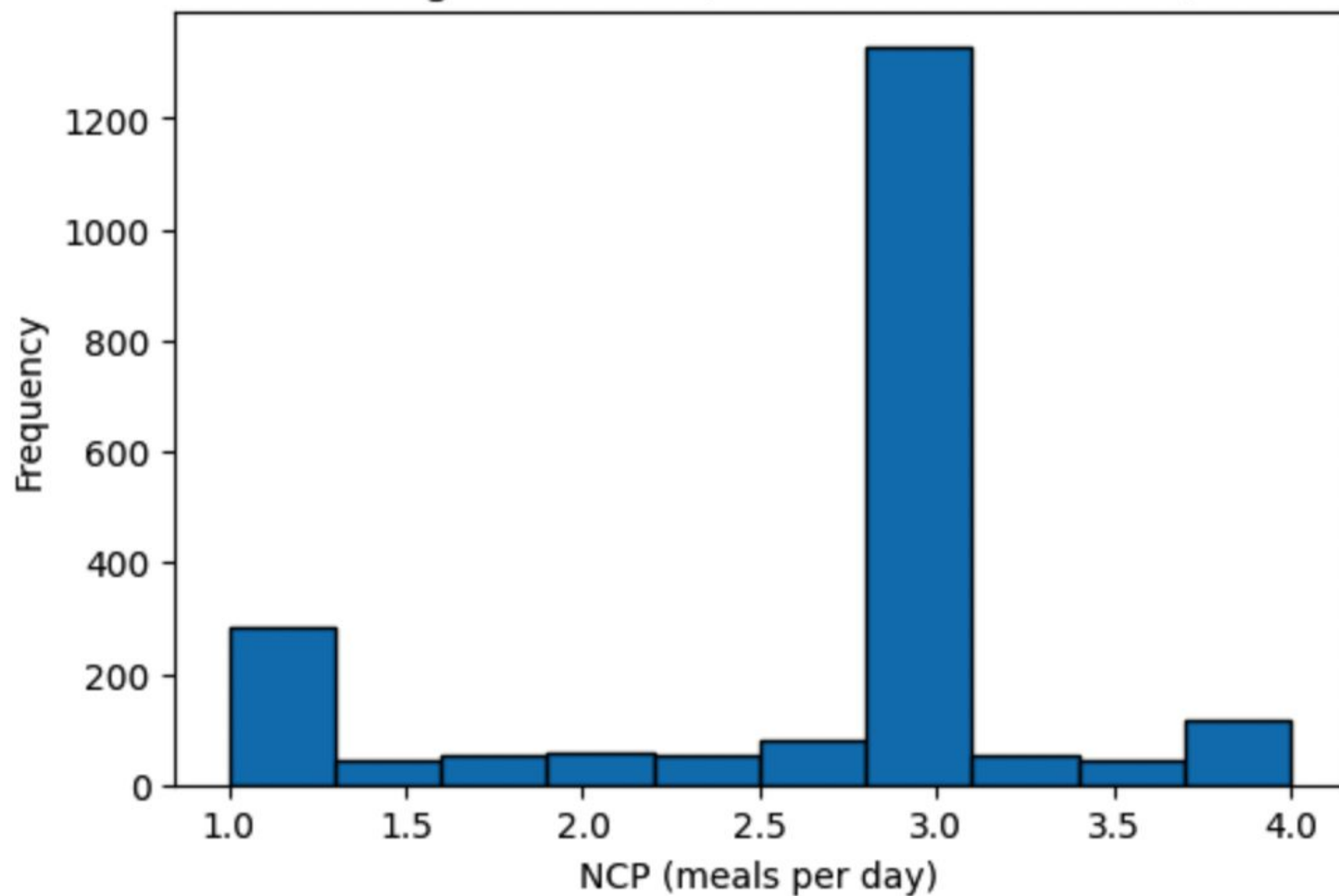
Binary Outcome Distribution (OverObese):

count

OverObese

1.0	1552
0.0	559

Histogram of NCP (Number of Main Meals)



Descriptive Statistics for NCP_num (meals per day):

NCP_num

count	2111.000000
--------------	-------------

mean	2.685628
-------------	----------

std	0.778039
------------	----------

min	1.000000
------------	----------

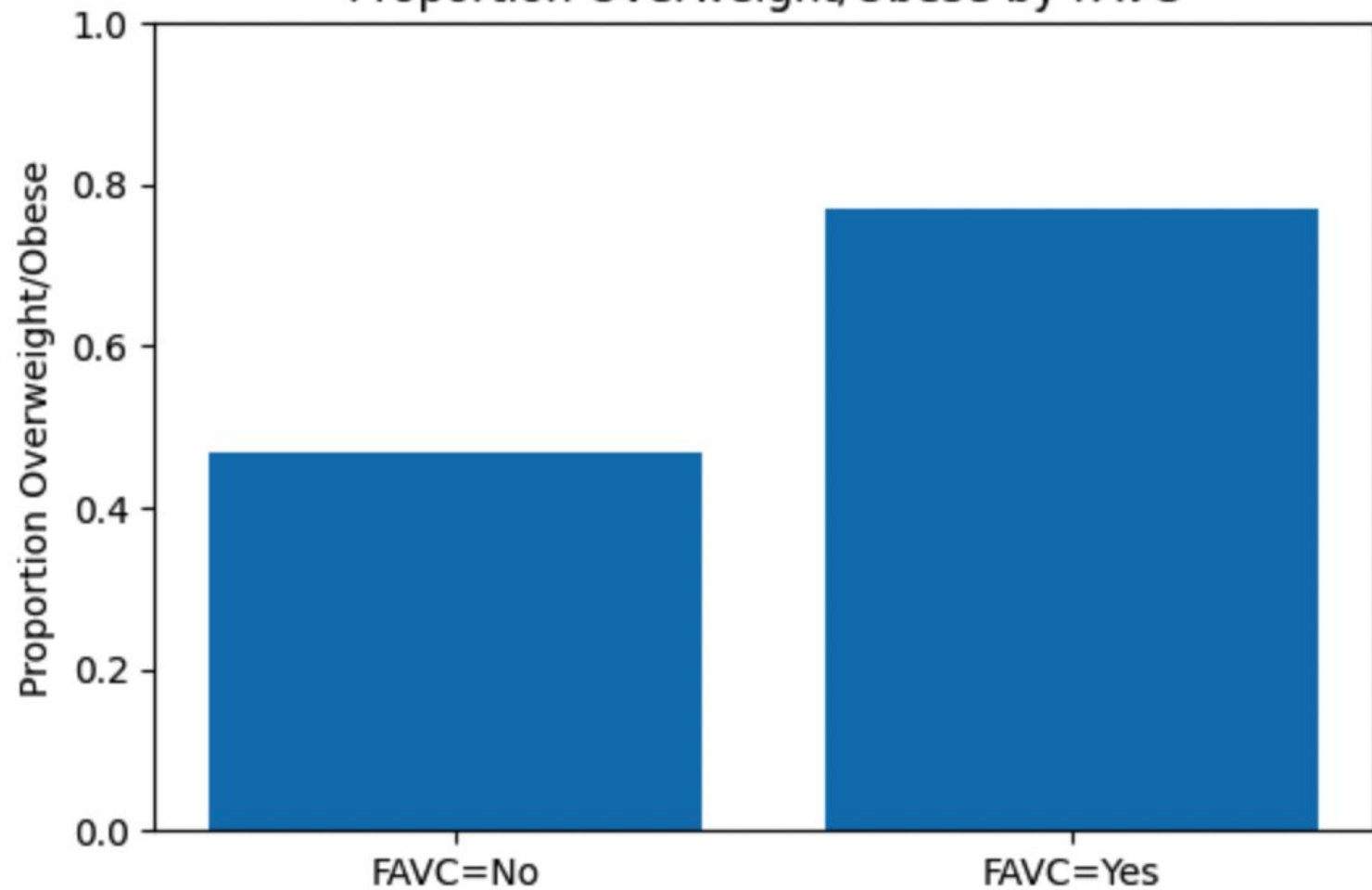
25%	2.658738
------------	----------

50%	3.000000
------------	----------

75%	3.000000
------------	----------

max	4.000000
------------	----------

Proportion Overweight/Obese by FAVC



Contingency Table: FAVC vs OverObese:

	Not Over/Obese	Over/Obese	Row Total
FAVC=No	130	115	245
FAVC=Yes	429	1437	1866
Total	559	1552	2111



Exploratory Data Analysis

Key Patterns:

- NCP Distribution: Heavily concentrated at 3 meals/day, limiting predictive strength
- Strong descriptive pattern by FAVC:
 - FAVC = Yes $\rightarrow$ ~77% overweight/obese
 - FAVC = No $\rightarrow$ ~47% overweight/obese
- Cross-tab suggests high-calorie food consumption correlates strongly with obesity before modeling
- Initial evidence supports that what you eat may matter more than how many times you eat

Hypothesis 1: Lifestyle Factors - Diet and Eating

- FAVC_bin (High Calorie Food Intake)
 - Coefficient = 1.9587
 - Odds Ratio = 7.09
 - p-value = 0.000078 → Highly significant
 - Interpretation: People who frequently eat high-calorie food are ~7× more likely to be overweight/obese
- NCP_num (Meal frequency):
 - Coefficient = -0.169
 - p-value = 0.2755 → NOT significant
 - Interpretation: Number of meals per day does not predict obesity
- Interaction (FAVC × NCP):
 - p-value = 0.197 → Not significant
- Overall evidence: Only high-calorie intake is statistically meaningful

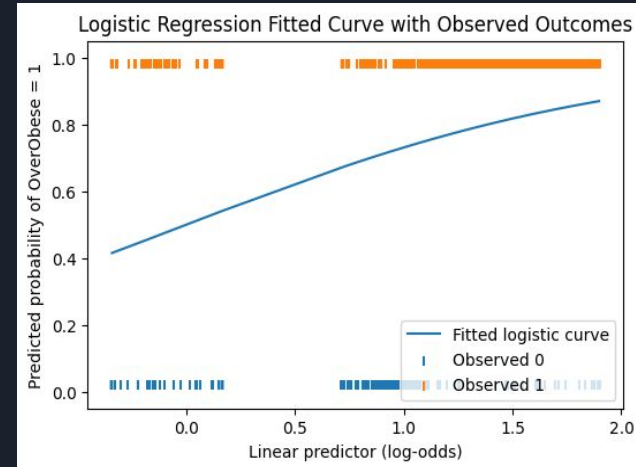
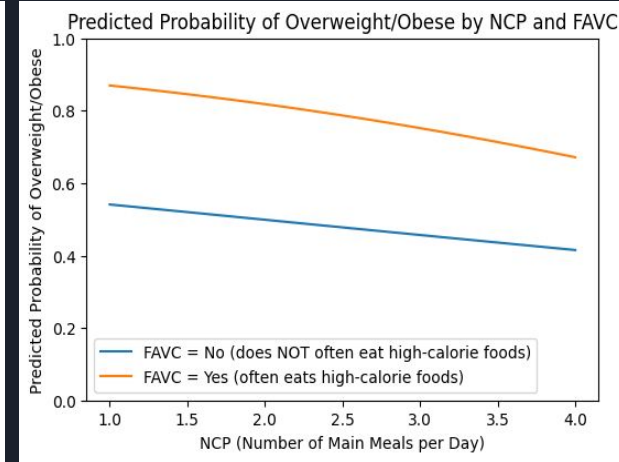
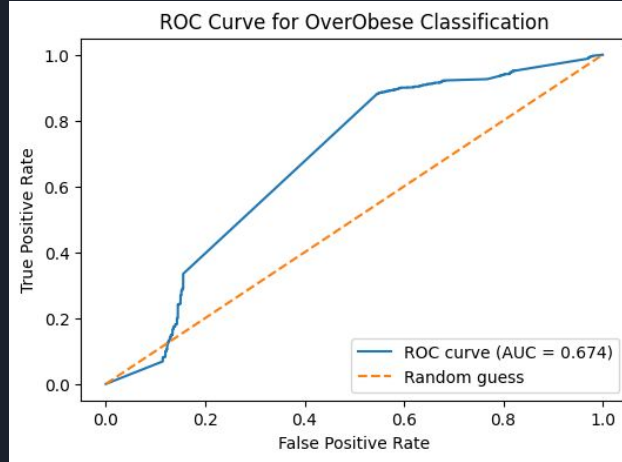
	Coef.	Std.Err.	z	P> z	[0.025	0.975]
Intercept	0.3333	0.4372	0.7624	0.4458	-0.5236	1.1903
FAVC_bin	1.9587	0.4957	3.9517	0.0001	0.9872	2.9302
NCP_num	-0.1690	0.1549	-1.0905	0.2755	-0.4726	0.1347
FAVC_bin:NCP_num	-0.2254	0.1747	-1.2901	0.1970	-0.5679	0.1171

ODDS RATIOS WITH 95% CI:

	Odds_Ratio	CI_lower	CI_upper	p_value
Intercept	1.395629	0.592389	3.288009	0.445810
FAVC_bin	7.090404	2.683811	18.732257	0.000078
NCP_num	0.844536	0.623349	1.144208	0.275483
FAVC_bin:NCP_num	0.798172	0.566703	1.124184	0.197024

Hypothesis 1: Lifestyle Factors - Diet and Eating Frequency

- ROC AUC = 0.674 → Model has moderate discrimination
- Predicted probability curves show:
 - FAVC = Yes → High predicted obesity risk (~0.80–0.90)
 - FAVC = No → Lower predicted risk (~0.45–0.55)
- Fitted logistic curve vs observed outcomes:
 - Shows good alignment with actual obesity classifications
 - Still realistic overlap due to behavioral/lifestyle variability
- Confirms statistical results: Model prediction is driven almost entirely by FAVC



Hypothesis 1: Lifestyle Factors - Diet and Eating Frequency

- Eating high-calorie foods matters a lot
 - Very strong effect ($p < 0.001$)
 - People who often eat high-calorie foods are much more likely to be overweight/obese
- Number of meals does NOT matter
 - NCP is not significant ($p = 0.28$)
 - Eating 1, 2, 3, or 4 meals a day doesn't change obesity risk much
- No interaction effect
 - Meal frequency does not change the impact of eating high-calorie foods
- Overall:
 - What you eat matters more than how often you eat
 - High-calorie food intake is the key predictor, not meal count



Hypothesis 2: Physical Activity and Transportation

Hypothesis 2: Individuals with lower physical activity frequency (FAF) and greater use of sedentary transportation modes (e.g., automobile or public transportation vs. walking/biking) have significantly higher BMI.

Dataset: UCI Obesity Levels (n=2,111)

Variables:

- **BMI** (calculated using Height and Weight)
- **FAF** (Physical Activity Frequency)
- **MTRANS** (Mode of Transportation)



Hypothesis 2: Physical Activity and Transportation

To evaluate Hypothesis 2, we used two complementary methods:

- One-way ANOVA: BMI differences across transportation modes (MTRANS).
- Simple linear regression: BMI regressed on physical activity frequency (FAF).



Hypothesis 2: Physical Activity and Transportation

- **ANOVA Model**

$$BMI_{ij} = \mu + \alpha_i + \varepsilon_{ij}, \quad \varepsilon_{ij} \sim N(0, \sigma^2)$$

Where μ is the overall BMI mean, α_i is the effect of the i -th transportation model (Automobile, Walking etc) and ε is the error term

- **Regression Model**

$$BMI_i = \beta_0 + \beta_1 FAF_i + \varepsilon_i, \quad \varepsilon_i \sim N(0, \sigma^2)$$

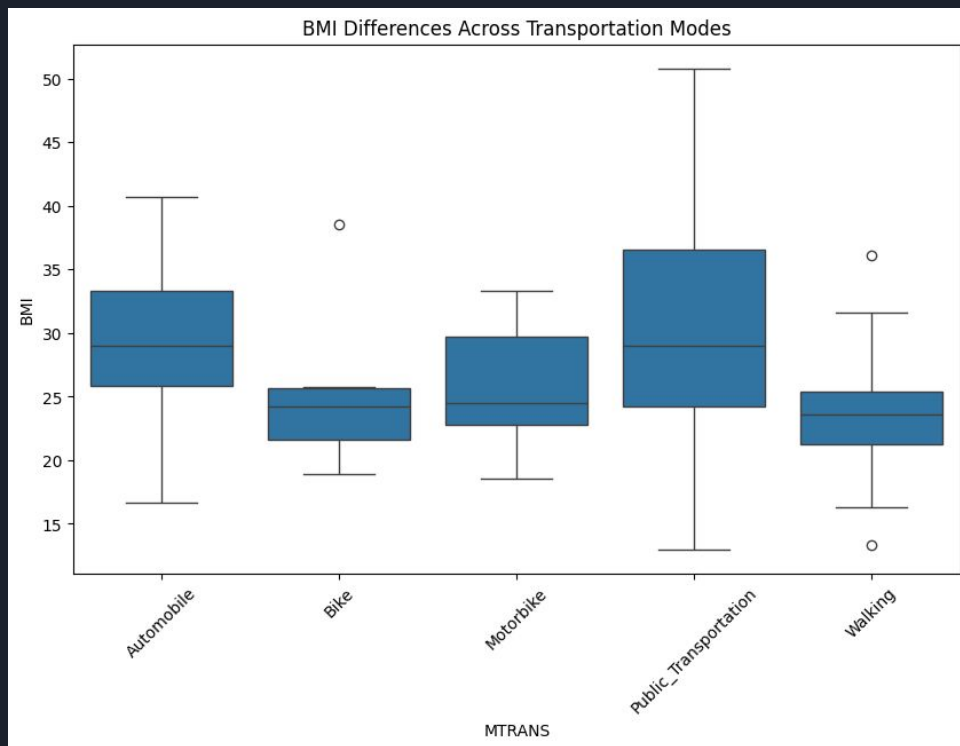
Where β_0 is the intercept, β_1 is the slope and ε is the error term.



Assumptions for ANOVA Model

- **Independence** - Each observation must be independent (no repeated measurements).
- **Normality** - Residuals within each group should be approximately normally distributed.
- **Homogeneity of variances** - All groups should have similar variance (equal spread).
- **Continuous outcome variable** - The dependent variable (BMI) must be numeric and continuous.

BMI differences across Transportation Modes





Hypothesis 2: Physical Activity and Transportation

ANOVA Table

```
=== ANOVA: BMI ~ Transportation Mode ===
```

	sum_sq	df	F	PR(>F)
C(MTRANS)	2740.645337	4.0	10.87522	9.968810e-09
Residual	132682.349877	2106.0	NaN	NaN

Since ANOVA p-value < 0.05 , Transportation mode significantly affects BMI



Linear Regression Assumptions

- Linearity: BMI has a linear relationship with FAF.
- Independence: Observations are independent of one another.
- Homoscedasticity: The variance of residuals is constant across FAF.
- Normality of residuals: The errors follow a normal distribution.

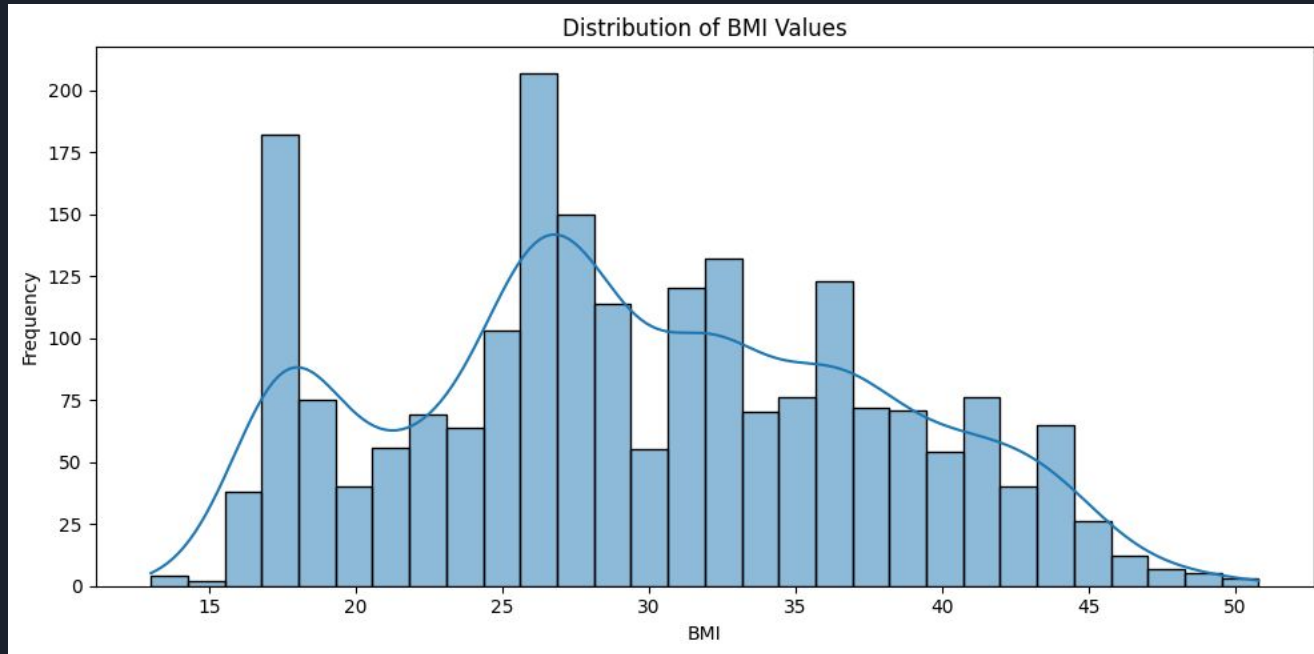


Hypothesis 2: Physical Activity and Transportation

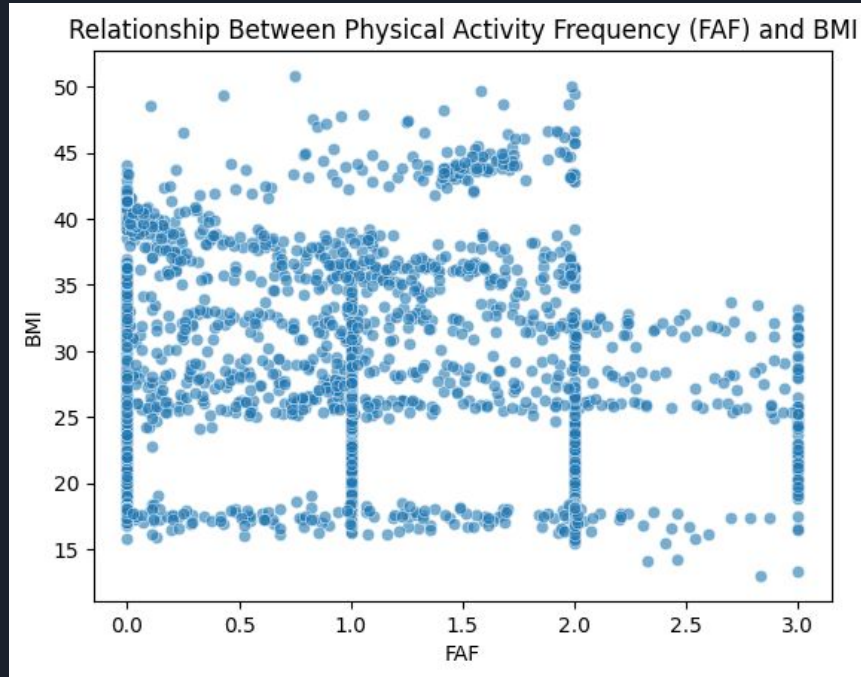
- Compute BMI given the data values using the formula: $BMI = Weight / (Height^2)$

```
# Computing BMI
df["BMI"] = df["Weight"] / (df["Height"] ** 2)
```

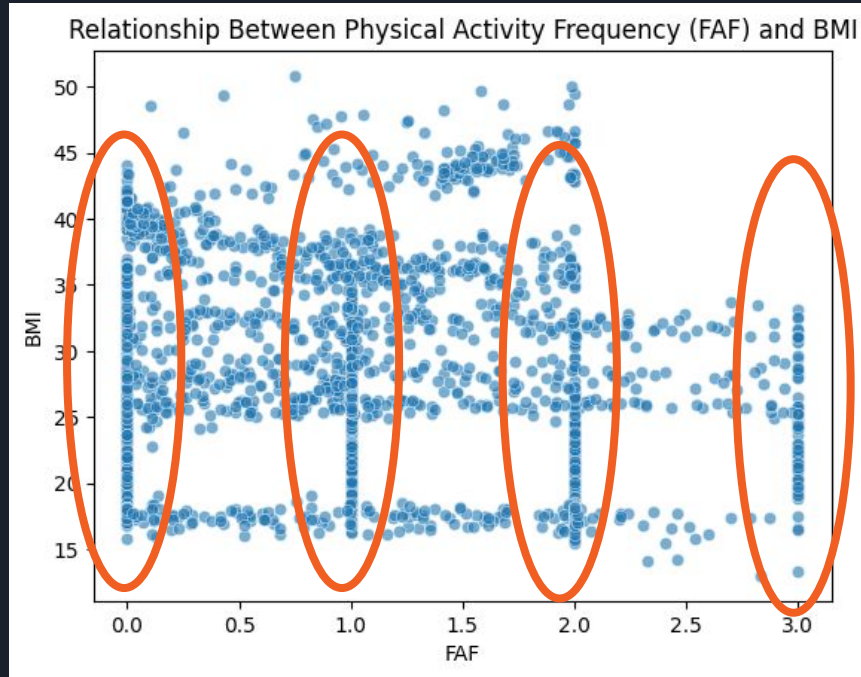
Hypothesis 2: Physical Activity and Transportation



Scatterplot of BMI against FAF



Scatterplot of BMI against FAF



Raw Data

Raw Data Preview (from ucimlrepo):

	Gender	Age	Height	Weight	family_history_with_overweight	FAVC	FCVC	NCP	CAEC	SMOKE	CH2O	SCC	FAF	TUE	CALC	MTRANS	NObeyesdad
0	Female	21.0	1.62	64.0	yes	no	2.0	3.0	Sometimes	no	2.0	no	0.0	1.0	no	Public_Transportation	Normal_Weight
1	Female	21.0	1.52	56.0	yes	no	3.0	3.0	Sometimes	yes	3.0	yes	3.0	0.0	Sometimes	Public_Transportation	Normal_Weight
2	Male	23.0	1.80	77.0	yes	no	2.0	3.0	Sometimes	no	2.0	no	2.0	1.0	Frequently	Public_Transportation	Normal_Weight
3	Male	27.0	1.80	87.0	no	no	3.0	3.0	Sometimes	no	2.0	no	2.0	0.0	Frequently	Walking	Overweight_Level_I
4	Male	22.0	1.78	89.8	no	no	2.0	1.0	Sometimes	no	2.0	no	0.0	0.0	Sometimes	Public_Transportation	Overweight_Level_II

Shape: (2111, 17)

Most FAF data points have integer values

OLS Regression Results

=== Regression: BMI ~ FAF ===

OLS Regression Results

```
=====
Dep. Variable:          BMI    R-squared:                0.032
Model:                  OLS    Adj. R-squared:             0.031
Method:                 Least Squares    F-statistic:          68.64
Date:                  Mon, 17 Nov 2025    Prob (F-statistic):      2.08e-16
Time:                  16:50:00    Log-Likelihood:         -7353.8
No. Observations:      2111    AIC:                   1.471e+04
Df Residuals:          2109    BIC:                   1.472e+04
Df Model:               1
Covariance Type:       nonrobust
=====
```

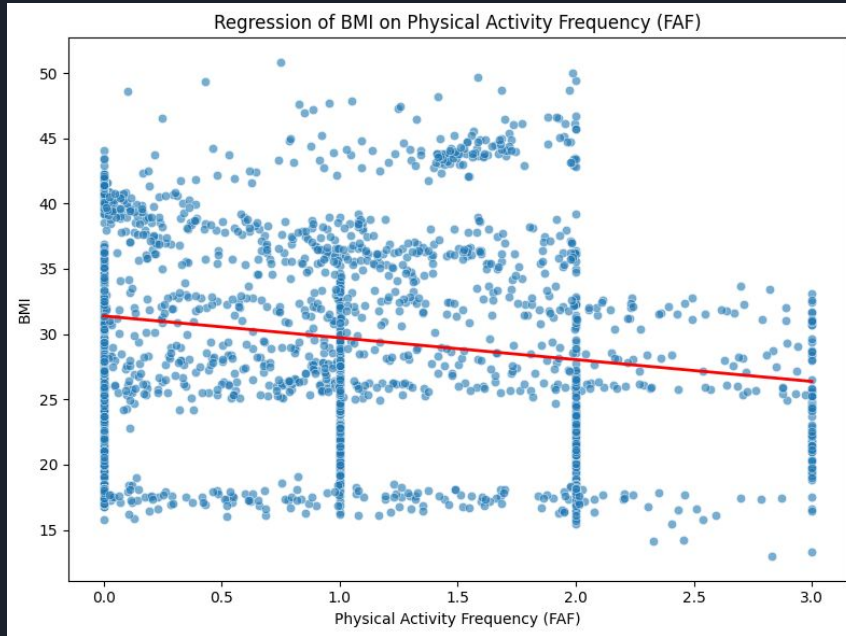
	coef	std err	t	P> t	[0.025	0.975]
Intercept	31.3895	0.267	117.771	0.000	30.867	31.912
FAF	-1.6721	0.202	-8.285	0.000	-2.068	-1.276

```
=====
Omnibus:                101.960    Durbin-Watson:          0.221
Prob(Omnibus):          0.000    Jarque-Bera (JB):       49.523
Skew:                   0.177    Prob(JB):               1.76e-11
Kurtosis:               2.338    Cond. No.               2.88
=====
```

- Intercept : 31.4
- FAF Coefficient : -1.67
- $P < 0.001$
- $R = -0.17$

Increasing physical activity frequency by one unit decreases BMI by approximately 1.67

Hypothesis 2: Physical Activity and Transportation



$$\text{BMI} = 31.39 - 1.67 \text{ FAF}$$



Conclusion

- For Hypothesis 1, frequent high-calorie food consumption is a strong predictor of being overweight or obese, whereas meal frequency alone is not. This suggests that what people eat matters more than how many times per day they eat.
- For Hypothesis 2, physical inactivity and sedentary transportation are linked to higher BMI. Even relatively simple measures like physical activity frequency (FAF) and Mode of Transportation capture meaningful differences in obesity risk.